Dhruv

Navigating The Future

Dhruv

Project ID : D001

Business Requirements Document

Unbound – College Event Management Platform

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Revision History

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1 Executive Summary

Unbound is a centralized event management platform developed specifically for MITAOE (MIT Academy of Engineering) that simplifies event organization, registration, and student participation. It addresses fragmentation caused by disparate tools like WhatsApp, spreadsheets, and email, providing one unified system for the college.

The platform aims to reduce manual coordination effort, increase student engagement, and improve visibility into event performance within MITAOE.

Key Benefits

- Centralized event management for MITAOE and its clubs
- Streamlined student registration experience
- Real-time analytics and performance tracking
- Reduced administrative overhead for club coordinators

2 Project Objectives

Objective ID	Business Objective	Specific	Measurable	Achievable	Relevant
OBJ-01	**MVP Deployment**	Launch core modules: Admin, Club, Student, and Payments.	100% of "In-Scope" features live and functional.	5-member team focusing on core MVP functionality.	Solves the fragmentation of tools at MITAOE.
OBJ-02	**Club Onboarding**	Transition campus clubs from WhatsApp to Unbound.	15–20 clubs actively managing event profiles.	Direct engagement with current club coordinators.	Centralization requires high club participation.
OBJ-03	**Student Engagement**	Drive event discovery and registration through the platform.	Reach a minimum of 2,000 unique registrations.	Targeted at the existing MITAOE student body.	Primary measure of platform value and success.
OBJ-04	**Payment Automation**	Securely process fees for paid workshops and fests.	95% successful payment completion rate.	Integration via Razorpay API and sandbox testing.	Eliminates manual tracking and cash handling.
OBJ-05	**Admin**	Reduce time	50% reduction	Through	Frees up

	Efficiency	spent on manual event coordination.	in manual data entry time.	automated dashboards and data exports.	coordinator time for better event quality.

3 Project Scope

3.1 In Scope

The following areas are in scope for this project :

1) MITAOE Admin Features

Admin login (single college - MITAOE)
Fest creation and management
View all club events
Event performance dashboard
Club management (approve/manage clubs).

2) Club Management

- Club profile setup
- Club admin login
- Event creation (standalone or under fest)
- Event editing and management
- View club-specific registrations

3) Event Management

- Event creation and editing by clubs
- Event categories (technical, cultural, sports, workshops, etc.)
- Event details (date, venue, eligibility, description)
- Event status management (draft, published, cancelled)
- Registration capacity management

4) Student Features (MITAOE Students)

- Student account creation and login
- Browse all MITAOE events
- Filter events by category, date, club
- Search functionality
- One-click event registration
- My Events dashboard

5) Payment Integration

- Payment for paid events
- Payment gateway integration (Razorpay)
- Support for free and paid events
- Payment confirmation and receipt
- Transaction history for students

3.2 Out of Scope

The following areas are out of scope for this project:

- Multi-college support (only MITAOE)
- Mobile applications (iOS/Android)
- Certificate generation
- Advanced reporting dashboards
- Third-party integrations (calendar sync, social media)
- SMS notifications
- QR code check-in system
- Marketing features
- College onboarding workflows

4 Business Requirement

BR ID	Requirement Title	Business Requirement	Key Subpoints (Important Only)	Business Justification
BR 1	MITAOE Admin System	Admin must manage the platform and oversee all club activities.	Create & manage fests- View all club events- Manage club profiles- View college-wide dashboard	Ensures centralized governance and institutional control.
BR 2	Club Management	Clubs must independently manage their events.	Create & manage events- Manage registrations- View club-level analytics	Empowers clubs while maintaining oversight.
BR 3	Event System	Events must be structured and informative.	Categorized events- Complete event details (date, venue, eligibility, capacity)- Event status tracking	Provides clarity and transparency for students.
BR 4	Student Experience	Students must easily discover and register for events.	Browse & filter events- Search functionality- One-click registration- My Events dashboard	Improves student participation and satisfaction.
BR 5	Authentication & Authorization	System must support secure role-based access.	Role-based access (College Admin, Club, Student)- Secure login & password recovery	Protects data and ensures proper access control.
BR 6	Data & Reporting	Platform must provide essential event metrics.	Registration tracking- Capacity monitoring- Basic report export	Enables data-driven decision making.
BR 7	Payment Processing	Platform must support online payments for paid events.	Free/Paid event option- Secure payment gateway integration- Transaction tracking	Eliminates manual payment handling and enables monetization.

5 Current Process

5.1.1 Current Process Description

Currently, event management at MITAOE is decentralized and relies on manual coordination across fragmented platforms. The process typically follows these steps:

1. **Planning:** Club coordinators plan events offline or via internal WhatsApp groups.
2. **Promotion:** Marketing is done via fragmented WhatsApp status updates and unofficial posters, leading to poor discoverability.
3. **Registration:** Students fill out disparate Google Forms; there is no single place to see all live events.
4. **Payment:** If an event is paid, coordinators manually verify screenshots of UPI transactions or collect cash.
5. **Tracking:** Data is manually moved from Google Forms to Excel Spreadsheets for attendance and record-keeping.
6. **Reporting:** No centralized reporting exists; college admins have no real-time visibility into club performance or student engagement.

5.1.2 Problem Statement

The absence of a unified digital infrastructure for MITAOE’s 15–20 clubs has resulted in an inefficient "siloed" environment. Key issues include:

- **Discovery Deficit:** Students frequently miss events because there is no central "source of truth" to browse upcoming activities.
- **Data Integrity Risks:** Manual tracking in spreadsheets leads to duplicate entries, lost payment records, and inconsistent data.
- **Operational Overload:** Club coordinators spend more time on administrative data entry and payment verification than on event quality.
- **Lack of Oversight:** MITAOE Administration cannot monitor college-wide event trends, participation rates, or financial transparency.

6 Proposed Process

The implementation of Unbound transforms the current fragmented workflow into a streamlined, automated, and centralized ecosystem. By integrating registration, payment, and management into a single platform, we eliminate manual data entry and provide a unified experience for all stakeholders.

6.1.1 Proposed Process Description

The new process centralizes all activities within the Unbound web platform, following these optimized steps:

1. **Centralized Planning:** Club coordinators create events directly on the Unbound portal, selecting categories and setting capacities.
2. **Unified Discovery:** Students visit the single "Live Events" dashboard to browse, search, and filter all upcoming activities across all 15–20 clubs.
3. **One-Click Registration:** Students register with a single click using their saved profiles, eliminating repetitive data entry.
4. **Automated Payment:** For paid events, the integrated Razorpay gateway handles transactions securely, instantly updating the registration status.
5. **Real-Time Analytics:** Attendance lists and financial reports are generated automatically. Both Club Coordinators and MITAOE Admins can view performance metrics instantly.
6. **Institutional Oversight:** The college administration monitors all club activity from a high-level dashboard to ensure quality and compliance.

6.1.2 Key Improvements

Feature	Current Process (Manual)	Proposed Process (Unbound)
Event Visibility	Scattered (WhatsApp/Posters)	**Centralized** (Unified Web Portal)
Registration	Google Forms (Manual Entry)	**One-Click** (Profile-based)
Payment	Manual UPI Screenshots/Cash	**Automated** (Razorpay Integration)
Data Management	Manual Spreadsheets	**Live Database** (Real-time tracking)
Admin Oversight	None (Siloed Clubs)	**Global Dashboard** (Full Visibility)

6.1.3 Process Impact Analysis

- **Reduced Friction:** By removing the need for manual payment verification and multiple form-fills, the "time-to-register" is reduced by over 80%.
- **Data Integrity:** A single database ensures no duplicate entries and accurate financial reconciliation.
- **Increased Participation:** Improved discoverability ensures that events reach their target audience effectively, maximizing student engagement.
- **Standardization:** All clubs follow a professional, uniform process, enhancing the institutional brand of MITAOE.

6.1.4 Key Stakeholders

Stakeholder	Role / Interest	Key Responsibilities	Involvement Level
MITAOE Admin	**Platform Governance** & Oversight	Approving club profiles, monitoring college-wide analytics, and ensuring event quality standards.	**High**
Club Coordinators	**Event Operations** & Execution	Creating/managing events, tracking registrations, and managing club-specific dashboards.	**High**
MITAOE Students	**End Users** & Participants	Browsing events, performing one-click registrations, and completing secure payments.	**High**
Development Team	**Technical Delivery**	System architecture, frontend/backend development, and API integration (Razorpay).	**High**
QA / Testing Team	**Quality Assurance**	Conducting UAT (User Acceptance Testing), bug tracking, and performance validation.	**Medium**
Finance/Accounts Office	**Financial Audit**	Overseeing automated payout reconciliations from the Razorpay gateway.	**Low**

7 Constraints & Assumptions

This section outlines the underlying factors that must remain true for the project to succeed, as well as the boundaries that limit the development of the **Unbound** platform.

7.1.1 Project Constraints

The following limitations define the boundaries of the **Unbound MVP**:

Category	Constraint Details
Scope	Development is strictly limited to **MITAOE** (Single-college architecture). No marketing or user acquisition activities are included.
Technical	No native mobile apps (iOS/Android) for Version 1.0; accessible via **web browsers only**. Payment integration is limited to **Razorpay**.
Resource	Project must be delivered by a **5-member team** with no additional budget for external sales or marketing resources.
Time	**30-day (1 month) hard deadline** for MVP launch, requiring parallel development and a "feature-freeze" on non-essential items.
Regulatory	Must comply with institutional data privacy policies and ensure secure storage of student personally identifiable information (PII).

7.1.2 Project Assumptions

The project team proceeds based on the following assumptions:

- **Infrastructure:** MITAOE students and staff have consistent internet access via mobile or desktop devices.
- **Institutional Support:** MITAOE administration fully supports the transition from manual to digital event management.
- **Participation:** At least 15–20 active college clubs will adopt the platform as their primary management tool.
- **Resource Availability:** Club coordinators can allocate time for initial training and data migration.
- **User Preference:** Students prefer a streamlined digital registration process over manual forms and WhatsApp coordination.

8 Success Criteria & KPIs (Key Performance Indicators)

The success of the **Unbound** platform will be measured by its ability to meet specific operational, engagement, and technical benchmarks during the first academic semester.

8.1.1 Operational Metrics

- **Launch Readiness:** 100% deployment of the web platform within the **1-month** project window.
- **Registration Volume:** Facilitate **2,000+** total event registrations across all college activities in the first semester.
- **Content Volume:** Onboard at least **100 individual events** created by the participating clubs.

8.1.2 Engagement & Adoption Metrics

- **Student Satisfaction:** Achieve a satisfaction score of ≥ **85%** based on post-registration user surveys.
- **Participation Growth:** Target a **30% increase** in overall student event attendance due to improved discoverability.
- **Club Adoption Rate:** Ensure **70% or more** of active MITAOE clubs move their registration process exclusively to Unbound.

8.1.3 System Performance Metrics

- **Platform Reliability:** Maintain a system uptime of **99%** throughout the academic year.
- **User Experience:** Average page load time must remain **under 3 seconds** to ensure low bounce rates.
- **Funnel Efficiency:** Achieve a registration completion rate of **> 90%** (from clicking "Register" to confirmation).

9 Risks & Mitigation Strategies

This section identifies potential threats to the project's success and provides a proactive plan to manage or eliminate them.

Risk	Impact	Probability	Mitigation Strategy
Aggressive Timeline	High	High	Use **Agile methodology** with daily standups and parallel frontend/backend development.
Payment Gateway Delays	High	Medium	Start Razorpay integration in **Week 1**; use sandbox environments for testing early.
Low Club Adoption	High	Medium	Host a **mandatory training session** for club coordinators and provide a "Quick-Start" guide.
Technical Bottlenecks	High	High	Focus strictly on **MVP features**; defer secondary requests to Version 2.0.
Security Breach	High	Low	Implement **JWT authentication**, conduct vulnerability audits, and use SSL encryption.
Scope Creep	High	High	Implement a **strict Change Request (CR) process**; any new features are added to the backlog for post-launch.
Team Unavailability	Medium	Medium	Maintain **thorough documentation** and cross-train developers to handle both frontend and backend tasks.

10 Glossary

Term	Explanation
MITAOE	MIT Academy of Engineering, the specific educational institution for which this platform is developed.
MVP	**Minimum Viable Product**: A version of a product with just enough features to be usable by early customers who can then provide feedback for future product development.
Siloed Environment	A situation where different groups (clubs) work independently and do not share information, leading to fragmentation.
Razorpay	The third-party payment gateway service used to process online transactions for paid events securely.
RBAC	**Role-Based Access Control**: A method of restricting system access to authorized users (Admin, Club, or Student) based on their specific role.
UAT	**User Acceptance Testing**: The final phase of the software testing process where actual users (students/coordinators) test the system to ensure it handles required tasks in real-world scenarios.
Fest	A large-scale college festival consisting of multiple technical, cultural, or sports events grouped under a single umbrella.
SaaS	**Software as a Service**: A software distribution model where a provider hosts applications; noted as "Out of Scope" as Unbound is currently a dedicated instance for MITAOE.
API	**Application Programming Interface**: A set of rules that allows the Unbound platform to communicate with external services like Razorpay.
Discoverability	The ease with which a student can find and learn about upcoming events within the platform.

11 References

Name	Link
MITAOE Official Website	https://mitaoe.ac.in/
Razorpay Standard Integration Docs	https://razorpay.com/docs/payments/payment-gateway/web-integration/standard/
Unbound Project GitHub Repository	https://github.com/PratikPBhosale/Unbound
Hack2Skill	https://hack2skill.com/
Unstop	https://unstop.com/

12 Appendix

12.1.1.1 Appendix A: User Stories

These stories define the functional requirements from the perspective of the end user to guide the development and testing phases.

User Role	Requirement (The "What")	Goal (The "Why")
Student	I want to filter events by category (Technical, Cultural, etc.).	So that I can quickly find events that align with my specific interests.
Student	I want to view my registration history in a personal dashboard.	So that I can keep track of upcoming event dates and venues in one place.
Club Coordinator	I want to export registration data to a CSV/Excel file.	So that I can manage attendance and logistics on the day of the event.
Club Coordinator	I want to set a "Registration Capacity" for my workshops.	So that I can prevent overbooking and ensure the quality of the session.
MITAOE Admin	I want to approve or reject a new Club's profile request.	So that I can ensure only authorized student bodies are hosting events on the platform.

12.1.1.2 Appendix B: Business Driver Analysis

A comparison of the impact of the **Unbound** platform against the current manual methods used at MITAOE.

- **Automation Driver:** Transitioning from manual UPI verification to Razorpay integration reduces the financial reconciliation cycle from **2 days to real-time**.
- **Centralization Driver:** Moving from 20+ fragmented WhatsApp groups to 1 unified portal increases event visibility by an estimated **300%** across the student body.
- **Data Accuracy Driver:** Eliminating Google Forms reduces the risk of duplicate data and "ghost registrations" by **90%**.